

Problem

Design an active first-order high-pass filter with

$$\mathbf{H}(s) = -\frac{100s}{s + 10}, \quad s = j\omega$$

Use a $1\text{-}\mu\text{F}$ capacitor.

Step-by-step solution

Step 1 of 1

For an active High pass filter

$$H(S) = -\frac{SC_i R_F}{1 + SC_i R_i}$$

$$\text{but } H(S) = -\frac{10s}{1 + \frac{s}{10}}$$

Comparing (1) & (2) leads to

$$C_i R_F = 10 \rightarrow R_F = 10 M\Omega$$

$$C_i R_i = 0.1 \rightarrow R_i = 100 K\Omega$$